

DDCO Laboratory (UE20CS256)

3rd Semester,
Date: 29/08/24

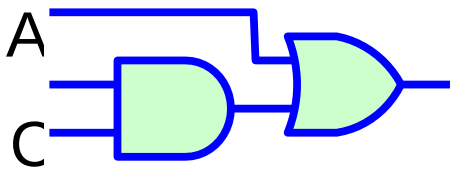
Academic Year 2024-25
Week-2 Submission

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Implement the following using the Verilog. After each experiment attach the following.

- Verilog Code Screenshot.
- Verilog VVP Output Screen Shot.
- GTKWAVE Screenshot.
- Output of Command Prompt

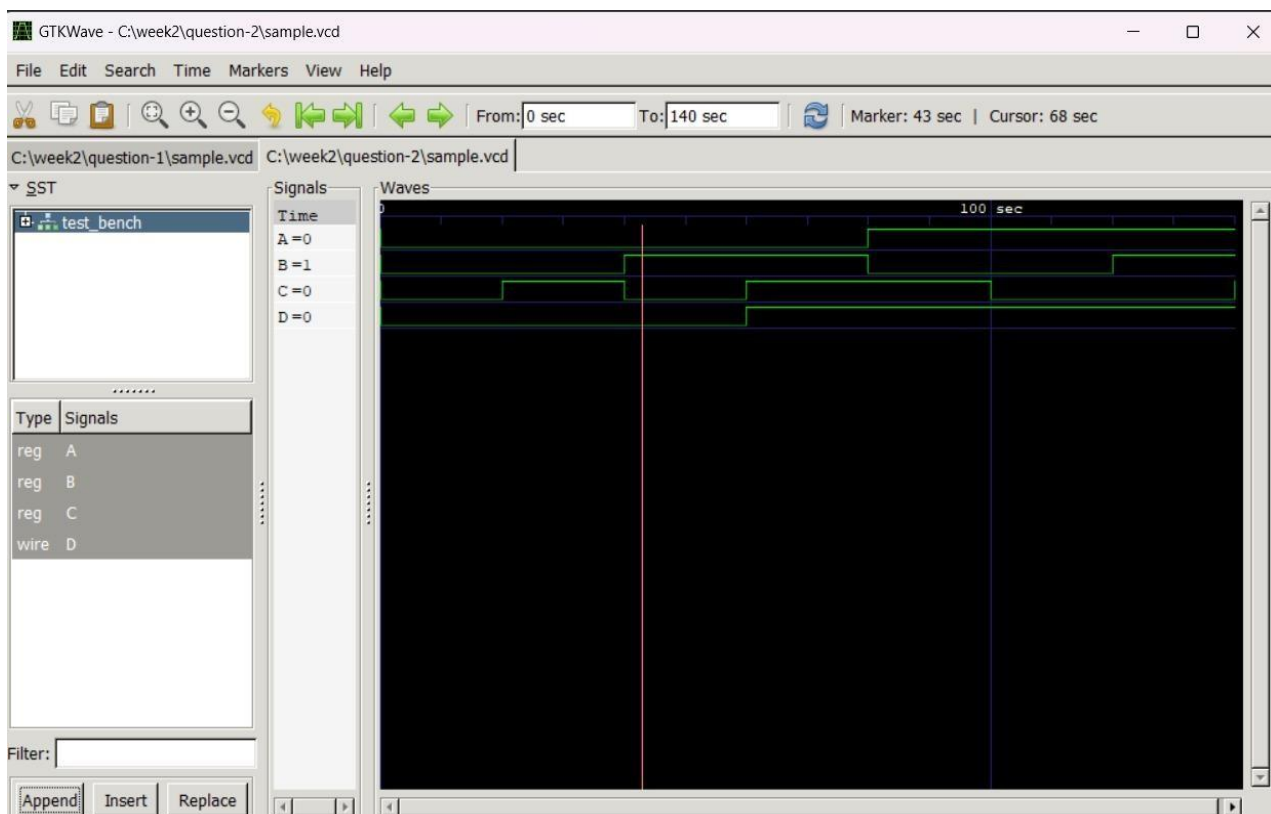
Circuit 1



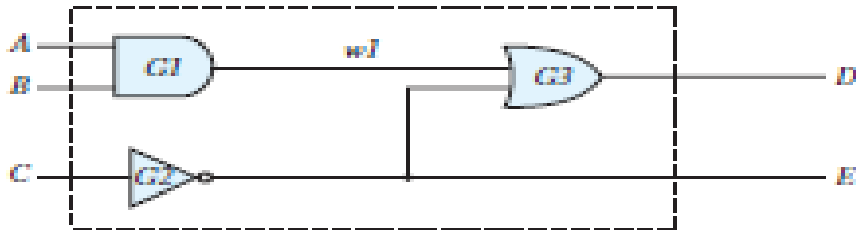
```
circuit.v  X  test_bench.v
question-2 > circuit.v
1 module circuit (A,B,C,D);
2   input A,B,C;
3   output D;
4   wire w1;
5
6   and G1(w1,B,C);
7   or G2(D,A,w1);
8 endmodule
```

```
circuit.v  test_bench.v X
question-2 > test_bench.v
1 module test_bench;
2
3   reg A,B,C;
4   wire D;
5
6   circuit M1(A,B,C,D);
7
8   initial begin
9     A = 0; B = 0; C = 0;
10    #20
11    A = 0; B = 0; C = 1;
12    #20
13    A = 0; B = 1; C = 0;
14    #20
15    A = 0; B = 1; C = 1;
16    #20
17    A = 1; B = 0; C = 1;
18    #20
19    A = 1; B = 0; C = 0;
20    #20
21    A = 1; B = 1; C = 0;
22    #20
23    A = 1; B = 1; C = 1;
24  end
25
26  initial begin
27    $monitor($time, "a = %b, b = %b, c = %b, d = %b", A, B, C, D);
28  end
29
30  initial begin
31    $dumpfile("sample.vcd");
32    $dumpvars(0, test_bench);
33  end
34 endmodule;
```

```
C:\week2\question-2>vvp test
VCD info: dumpfile sample.vcd opened for output.
      0a = 0, b = 0, c = 0, d = 0
      20a = 0, b = 0, c = 1, d = 0
      40a = 0, b = 1, c = 0, d = 0
      60a = 0, b = 1, c = 1, d = 1
      80a = 1, b = 0, c = 1, d = 1
     100a = 1, b = 0, c = 0, d = 1
     120a = 1, b = 1, c = 0, d = 1
     140a = 1, b = 1, c = 1, d = 1
```



Circuit 2



```

sample_circuit.v X test_bench.v
question-1 > sample_circuit.v
1 module sample_circuit (a,b,c,d,e);
2     input a,b,c;
3     output d,e;
4     wire w1;
5
6     and G1(w1,a,b);
7     not G2(e,c);
8     or G3(d,w1,e);
9 endmodule

```

```

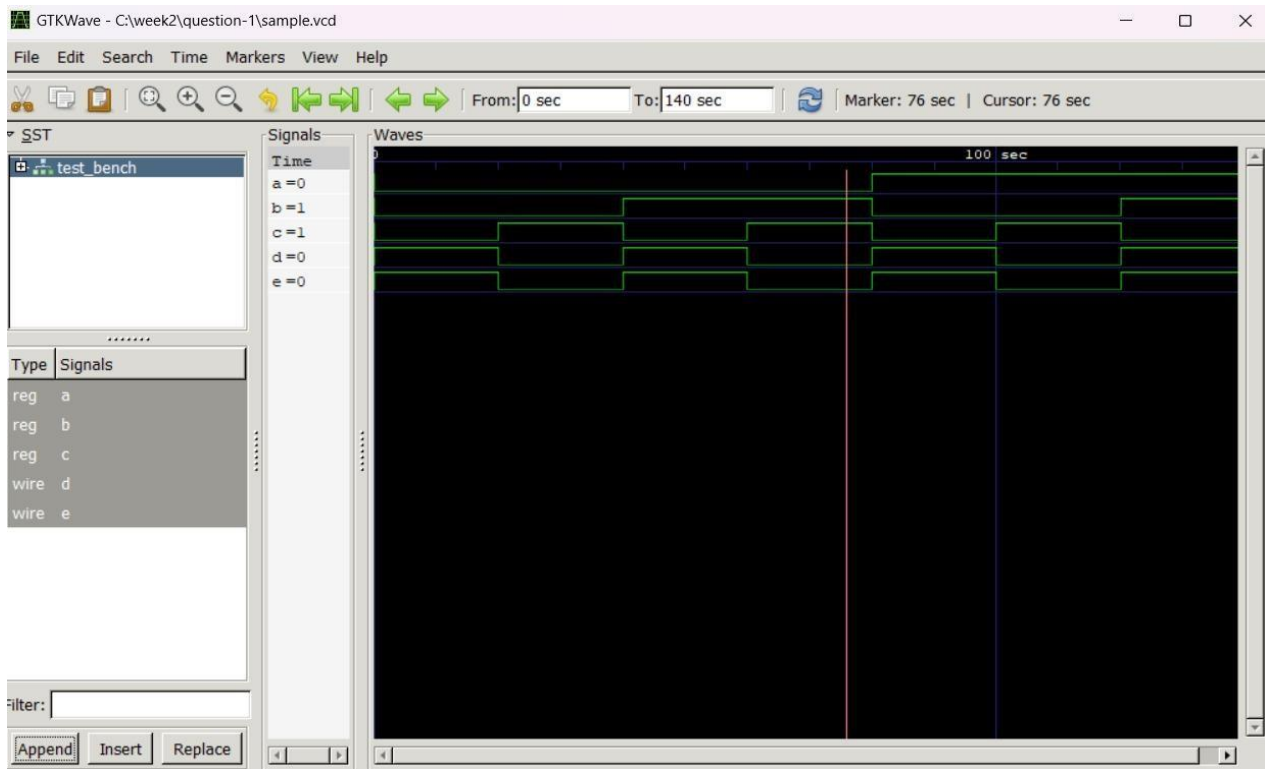
sample_circuit.v X test_bench.v
question-1 > test_bench.v
1 module test_bench;
2
3     reg a,b,c;
4     wire d,e;
5
6     sample_circuit M1(a,b,c,d,e);
7     initial begin
8         a = 0; b = 0; c = 0;
9         #20
10        a = 0; b = 0; c = 1;
11        #20
12        a = 0; b = 1; c = 0;
13        #20
14        a = 0; b = 1; c = 1;
15        #20
16        a = 1; b = 0; c = 0;
17        #20
18        a = 1; b = 0; c = 1;
19        #20
20        a = 1; b = 1; c = 0;
21        #20
22        a = 1; b = 1; c = 1;
23    end
24
25    initial begin
26        $monitor($time, "a = %b, b = %b, c = %b, d = %b, e = %b", a, b, c, d, e);
27    end
28
29    initial begin
30        $dumpfile("sample.vcd");
31        $dumpvars(0, test_bench);
32    end
33
34 endmodule

```

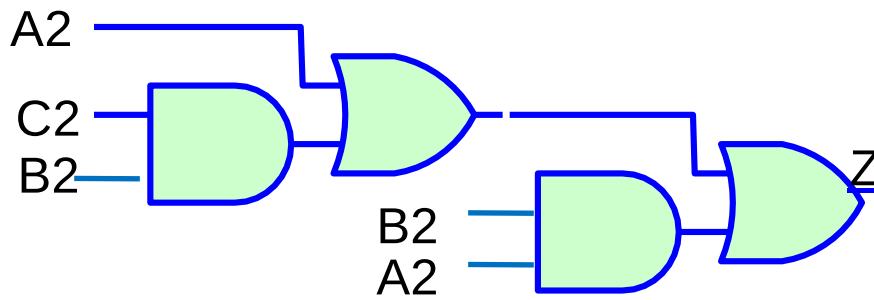
```

C:\week2\question-1>vvp test
VCD info: dumpfile sample.vcd opened for output.
    0a = 0, b = 0, c = 0, d = 1, e = 1
   20a = 0, b = 0, c = 1, d = 0, e = 0
   40a = 0, b = 1, c = 0, d = 1, e = 1
   60a = 0, b = 1, c = 1, d = 0, e = 0
   80a = 1, b = 0, c = 0, d = 1, e = 1
  100a = 1, b = 0, c = 1, d = 0, e = 0
  120a = 1, b = 1, c = 0, d = 1, e = 1
  140a = 1, b = 1, c = 1, d = 1, e = 0

```



Circuit 3



```
circuit.v  X  test_bench.v
question-3 > circuit.v
1 module circuit (A,B,C,D);
2
3     input A,B,C;
4     output D;
5
6     wire w1,w2,w3;
7
8     and G1(w1,B,C);
9     or G2(w2,w1,A);
10    and G3(w3,A,B);
11    or G4(D,w2,w3);
12 endmodule
```

```
circuit.v  test_bench.v X
question-3 > test_bench.v
1 module test_bench;
2
3     reg A,B,C;
4     wire D;
5
6     circuit M1(A,B,C,D);
7
8     initial begin
9         A = 0; B = 0; C = 0;
10        #20
11        A = 0; B = 0; C = 1;
12        #20
13        A = 0; B = 1; C = 0;
14        #20
15        A = 0; B = 1; C = 1;
16        #20
17        A = 1; B = 0; C = 1;
18        #20
19        A = 1; B = 0; C = 0;
20        #20
21        A = 1; B = 1; C = 0;
22        #20
23        A = 1; B = 1; C = 1;
24    end
25
26    initial begin
27        $monitor($time, "a = %b, b = %b, c = %b, d = %b", A, B, C, D);
28    end
29
30    initial begin
31        $dumpfile("sample.vcd");
32        $dumpvars(0, test_bench);
33    end
34 endmodule;
```

```
C:\week2\question-3>vvp test
VCD info: dumpfile sample.vcd opened for output.
      0a = 0, b = 0, c = 0, d = 0
     20a = 0, b = 0, c = 1, d = 0
     40a = 0, b = 1, c = 0, d = 0
     60a = 0, b = 1, c = 1, d = 1
     80a = 1, b = 0, c = 1, d = 1
    100a = 1, b = 0, c = 0, d = 1
    120a = 1, b = 1, c = 0, d = 1
    140a = 1, b = 1, c = 1, d = 1
```

